

ECE 3210 hw05

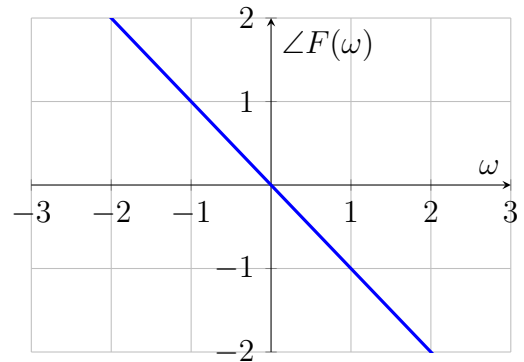
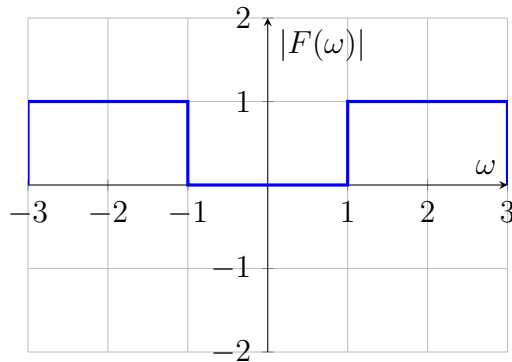
1. Find the Fourier transform of

$$f(t) = 2 \operatorname{sinc}(\pi t) \cos(\pi t)$$

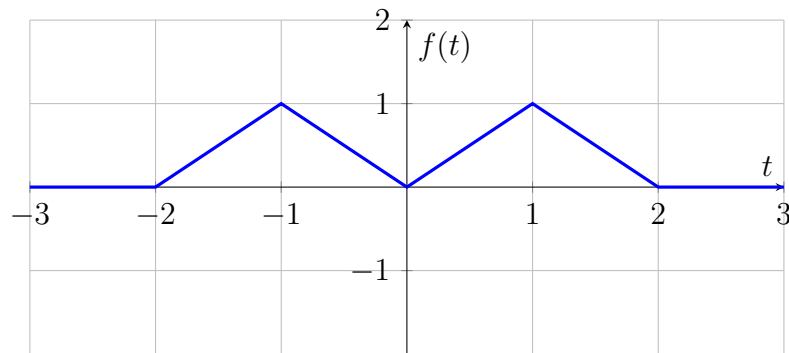
2. (a) Find the Fourier transform of

$$f(t) = \begin{cases} -t, & -2 \leq t \leq 0, \\ t, & 0 < t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

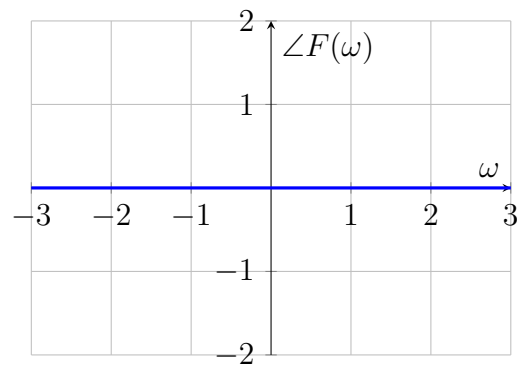
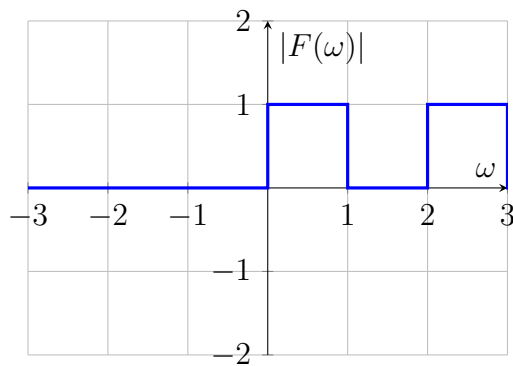
- (b) Consider the magnitude and phase shown below. Find the inverse transform $f(t)$.



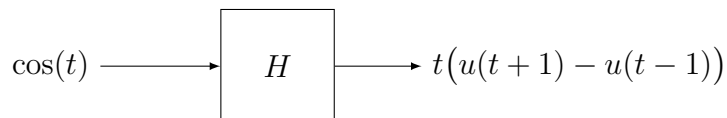
3. (a) Determine the Fourier transform of the waveform below.



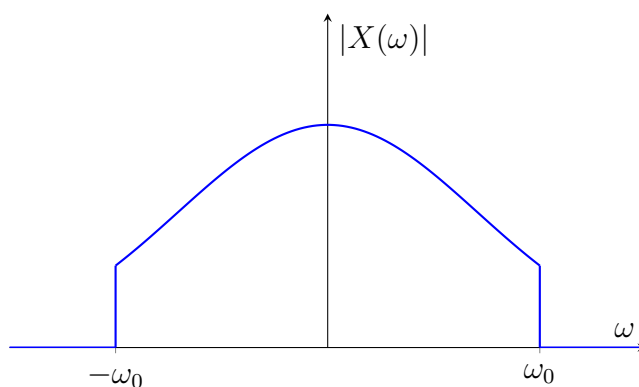
- (b) The magnitude and phase of a spectrum are supplied below. Find $f(t)$.



4. (a) For the input/output relationship below, find $Y(\omega)$. Check the symmetry of your final answer.



- (b) Is the system in part (a) linear and time invariant?
- (c) Let $x(t)$ be a signal whose spectrum is identically zero outside the range $-\omega_0 \leq \omega \leq \omega_0$. An example of such a spectrum is shown below. The actual values of $X(\omega)$ are not important for this problem, but the range of ω for which $X(\omega)$ is non-zero is.



For $y(t) = x(t) + x(t-1)$, if $X(\omega) = 0$ outside $[-\omega_0, \omega_0]$, state the possible nonzero range of $Y(\omega)$.

- (d) For $y(t) = x(t) \cos(\omega_0 t)$, if $X(\omega) = 0$ outside $[-\omega_0, \omega_0]$, state the possible nonzero range of $Y(\omega)$.

5. (a) Compute the integral

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1}{1 - j\omega} \right|^2 d\omega.$$

- (b) Compute the integral

$$\int_{-\infty}^{\infty} \cos(2\pi t) \operatorname{sinc}^2(t) \operatorname{sinc}(2t) dt.$$

- (c) Evaluate the integral

$$I = \int_{-\infty}^{\infty} \frac{2\operatorname{sinc}(t)}{t^2 + 1} dt$$

Answer Key

$$1 \quad F(\omega) = \text{rect}\left(\frac{\omega}{4\pi}\right)$$

$$2(a) \quad F(\omega) = 8 \text{sinc}(2\omega) - 4 \text{sinc}^2(\omega)$$

$$2(b) \quad f(t) = \frac{1}{\pi} [3 \text{sinc}(3(t-1)) - \text{sinc}(t-1)]$$

$$3(a) \quad F(\omega) = 4 \text{sinc}^2(\omega) - 2 \text{sinc}^2\left(\frac{\omega}{2}\right)$$

$$3(b) \quad f(t) = \frac{1}{2\pi} \left[3 \text{sinc}\left(\frac{3t}{2}\right) - \text{sinc}\left(\frac{t}{2}\right) \right] e^{j3t/2}$$

$$4(a) \quad Y(\omega) = \frac{2}{j\omega} [\text{sinc}(\omega) - \cos(\omega)]$$

$$4(b) \quad \text{The system is not LTI.}$$

$$4(c) \quad -\omega_0 \leq \omega \leq \omega_0$$

$$4(d) \quad -2\omega_0 \leq \omega \leq 2\omega_0$$

$$5(a) \quad E_Y = \frac{1}{2}$$

$$5(b) \quad 0$$

$$5(c) \quad I = 2\pi(1 - e^{-1})$$